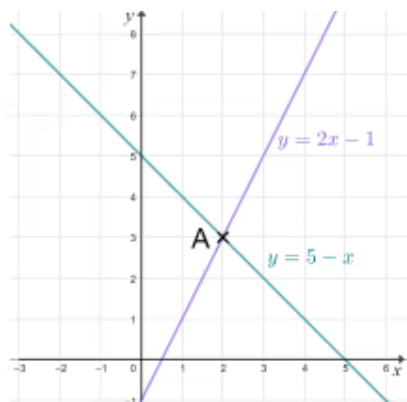




Problem solving with graphical representations

- 1 The diagram shows the lines $y = 2x - 1$ and $y = 5 - x$. The point A will help you find the solution to the _____ equations $y = 2x - 1$ and $y = 5 - x$. Tick **1** correct answer



- ☐ intersecting
☐ non-linear
☐ quadratic
☒ simultaneous

- 2 Sam plots the graphs of $y = 3x + 1$ and $y = 3x - 4$ on the same coordinate grid. How many points will lie on both lines? Fill in the blank

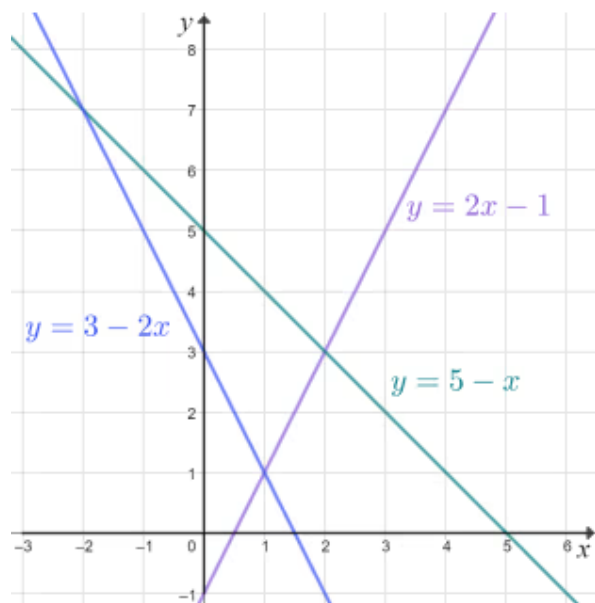
0

- 3 The linear equations $y = 2x - 9$ and $y = 3 - x$ intersect at the point with coordinates $(4, -1)$. What is the solution to the equation $2x - 9 = 3 - x$? Tick **0** correct answers

- ☐ $x = -1$
☐ $x = 4$
☐ $y = -1$
☐ $y = 4$

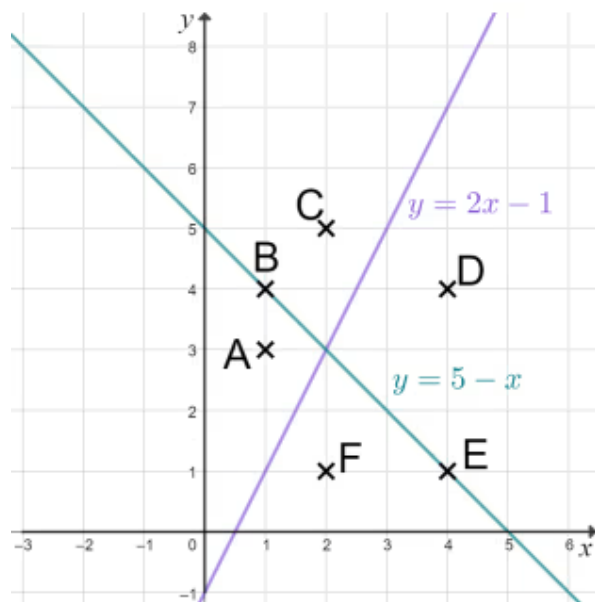


- 4 Use the graph to solve the simultaneous equations $y = 2x - 1$ and $y = 5 - x$. Tick **1** correct answer



- ☐ $x = 3, y = 2$
☐ $x = 7, y = -2$
☐ $x = 1, y = 1$
☐ $x = -2, y = 7$
☒ $x = 2, y = 3$

- 5 The diagram shows 6 points and the the lines $y = 2x - 1$ and $y = 5 - x$. Which points satisfy the inequality $y < 5 - x$? Tick **2** correct answers



- ☒ A
☐ C
☐ D
☒ F
☐ B and E



- 6 The diagram shows 6 points and the the lines $y = 2x - 1$ and $y = 5 - x$. The point that is the region $y > 5 - x$ and $y > 2x - 1$ is C. Fill in the blank

